

Experiment No-2

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Branch: MCA

Semester: 3rd

Subject Name: Business Analytics Lab

UID: 24MCA20045

Section/Group: 24MCA-1(B)

Date of Performance: 30/07/2025

Subject Code: 24CAH-703

1. Aim of the practical:

To learn how to create and utilize Pivot Tables in Excel to quickly summarize and analyze large datasets, including grouping data within Pivot Tables and creating multiple Pivot Tables on a single worksheet for comprehensive data analysis.

2. Task to be Done:

- Insert Pivot Table for car sales data of Group values based on Colour.
- Calculate Sum of Price and Count of Colour.
- Create Pivot Layout to compare various colours.
- Present aggregated results clearly

3. Apparatus:

- **Microsoft Excel (2016 or above):**
Primary tool for creating and formatting the worksheet using formulas.
- **Excel Dataset (Student Marks Sheet):**
Predefined data (Make, Model, Mileage, Price, Amount, Colour) to work on.

4. Procedure:

a) Creating a Pivot Table and Grouping Data:

1. Select any cell within your dataset (make sure your data has column headers)
2. Go to the Insert tab and click on PivotTable
3. In the dialog box, verify the data range and choose where to place the PivotTable (new worksheet or existing one)
4. Click OK to create a blank PivotTable

5. In the PivotTable Fields pane, drag fields to the appropriate areas (Rows, Columns, Values)
6. To group data, right-click on an item in the PivotTable, select Group, then specify your grouping criteria
(dates, numbers, etc.)
7. Use the Group Field dialog to set your grouping parameters and click OK

b) Creating Multiple Pivot Tables on One Worksheet:

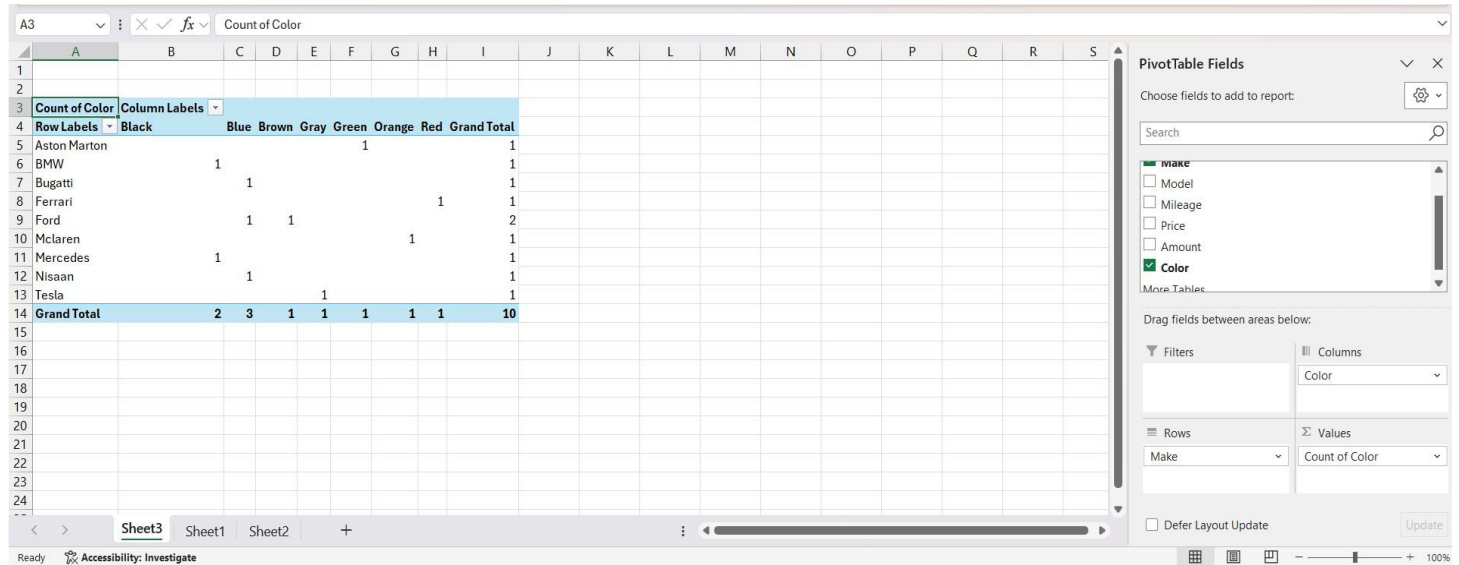
1. Create your first PivotTable as described above.
2. Click anywhere in your original dataset.
3. Insert another PivotTable (Insert > PivotTable).
4. This time, choose **Existing Worksheet** and select a location that doesn't overlap with your first PivotTable.
5. Configure this second PivotTable with different fields to show alternative views of your data.
6. Repeat the process for additional PivotTables, ensuring they don't overlap.
7. Use different layouts and styles to distinguish between multiple PivotTables.

5. Output:

a. Dataset:

	A	B	C	D	E	F	G
1	Sr. No	Make	Model	Milage	Price	Amount	Colour
2	1	Ford	Mustang	30	\$ 5,000	\$ 4,500	Red
3	2	Nissan	Rougue	40	\$ 6,000	\$ 5,500	Blue
4	3	Volkswgar	Jetta	28	\$ 3,000	\$ 2,500	Green
5	4	BMW	X5	30	\$ 4,300	\$ 4,000	Yellow
6	5	Ford	F-150	43	\$ 4,200	\$ 4,000	Purple
7	6	Nissan	Rougue	31	\$ 5,400	\$ 5,000	Black
8	7	Volkswgar	Jetta	29	\$ 3,900	\$ 3,500	Red
9	8	Ford	Mustang	26	\$ 2,900	\$ 2,500	Blue
10	9	Nissan	Rougue	33	\$ 3,100	\$ 3,000	Yellow
11	10	Volkswgar	Jetta	42	\$ 2,800	\$ 2,500	White

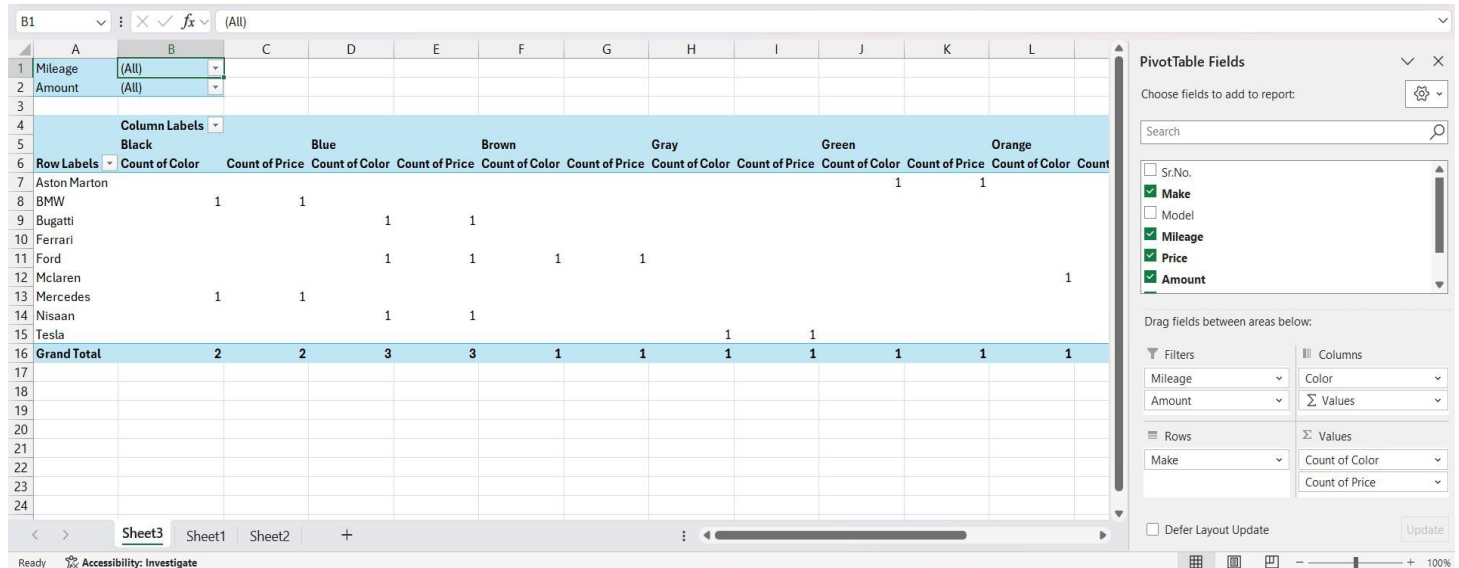
b. Drag Color to rows



Excel screenshot showing a PivotTable setup. The PivotTable is located in the range A3:S14. The PivotTable Fields task pane on the right shows 'Color' selected in the Values area and 'Make' selected in the Rows area. The PivotTable data is as follows:

Make	Blue	Brown	Gray	Green	Orange	Red	Grand Total
Aston Marton				1			1
BMW	1						1
Bugatti		1					1
Ferrari					1		1
Ford		1	1				2
McLaren				1			1
Mercedes	1						1
Nissan		1					1
Tesla				1			1
Grand Total	2	3	1	1	1	1	10

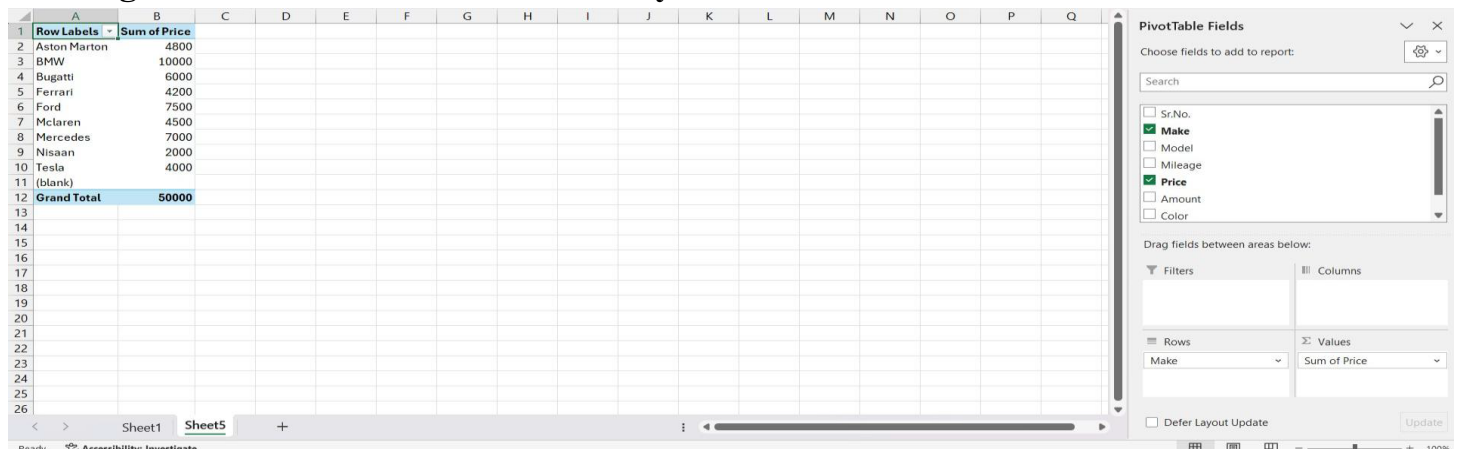
c. Filtering based on Mileage



Excel screenshot showing a PivotTable setup. The PivotTable is located in the range A6:L16. The PivotTable Fields task pane on the right shows 'Price' selected in the Values area and 'Make' selected in the Rows area. The 'Mileage' filter is applied, showing only cars with 'All' mileage. The PivotTable data is as follows:

Make	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price	Count of Price
Aston Marton												
BMW	1	1										
Bugatti			1	1								
Ferrari					1	1	1	1				
Ford												
McLaren												1
Mercedes	1	1										
Nissan				1	1							
Tesla							1	1				
Grand Total	2	2	3	3	1	1	1	1	1	1	1	1

d. Drag Price to Values → Summarize by SUM



Excel screenshot showing a PivotTable setup. The PivotTable is located in the range A2:Q12. The PivotTable Fields task pane on the right shows 'Price' selected in the Values area and 'Make' selected in the Rows area. The PivotTable data is as follows:

Make	Sum of Price
Aston Marton	4800
BMW	10000
Bugatti	6000
Ferrari	4200
Ford	7500
McLaren	4500
Mercedes	7000
Nissan	2000
Tesla	4000
(blank)	
Grand Total	50000

6. Learning outcomes (What I have learnt):

- Understand the purpose and benefits of using Pivot Tables for data analysis.
- Learn how to create Pivot Tables from large datasets efficiently.
- Gain the ability to summarize data using functions like Sum, Count, Average, etc.
- Master how to group data by date, number ranges, or categories within a Pivot Table.
- Learn to sort and filter Pivot Table data for more targeted analysis.

Evaluation Grid:

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Demonstration and Performance		12
2.	Worksheet		8
3.	Via		10